Threat-Intelligence Kill-Chain Correlation Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Threat-Intelligence Kill-Chain Correlation Engine is a cybersecurity system designed to analyze and correlate security events from multiple sources to identify complete attack sequences instead of isolated alerts. Modern networks generate thousands of security logs, making it difficult for analysts to detect real threats efficiently. This project collects security events, applies advanced data structures and algorithms such as Graphs, Tries, KMP, and Rabin-Karp for efficient pattern matching and event correlation, and reconstructs the attacker's progression through the cyber kill chain. The system visualizes attack paths, prioritizes threats based on severity, and provides real-time insights through an interactive dashboard. By reducing alert fatigue and improving threat detection accuracy, the proposed solution helps security teams respond faster to cyber incidents. The project demonstrates the practical application of data structures and algorithms in cybersecurity while providing a scalable foundation for future integration with IoT-based security monitoring systems and AI-driven threat intelligence.

Signature of the Guide